

14. **Ice Cream, Anyone?** The popular ice cream franchise *Coldstone Creamery* posted the nutritional information for its ice cream offerings in three serving sizes—"Like it," "Love it," and "Gotta Have it"—on their website.¹⁸ A portion of that information for the "Like it" serving size is shown in the table.

Flavor	Calories X	Total Fat (grams) y
Cake Batter	340	19
Cinnamon Bun	370	21
French Toast	330	19
Mocha	320	20
OREO® Crème	440	31
Peanut Butter	370	24
Strawberry Cheesecake	320	21

- a. Should you use the methods of linear regression analysis or correlation analysis to analyze the data? Explain.
- b. Analyze the data to determine the nature of the relationship between total fat and calories in *Coldstone Creamery* ice cream.

b. $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = 0.91896$

$H_0: \rho = 0, H_A: \rho > 0$

$\alpha = 0.05$

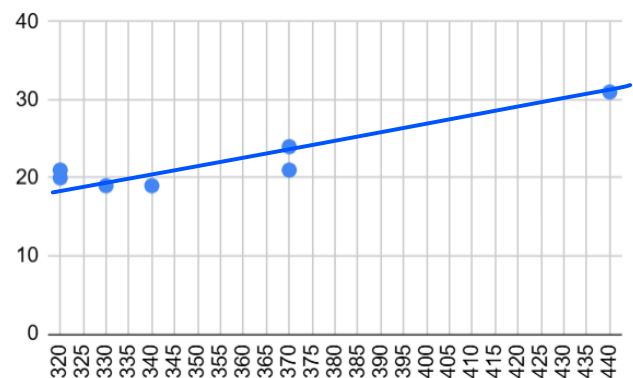
$t_{STAT} = r \sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2}$

$t^* = 0.91896 \sqrt{\frac{5}{1-0.91896^2}} = 5.2122$

$RR: \{t \mid |t| > t_{0.05, 5}\} = \{t \mid |t| > 2.015\}$

$5.2122 > 2.015 \Rightarrow \text{Reject } H_0$

$\Rightarrow$ There is strong positive relationship between X i. y .



Yes, because there seems to have positive relationship that, higher calories has higher total fat.